DATA 200 LAB 1 REPORT

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Project files	**checkmygrade.py, test_checkmygrade.py, students.csv, courses.csv, professors.csv, login.csv**
Report focus	Alignment with Lab 1 requirements, actual CSV records, and unit-test evidence

1. Project overview

This project creates a console-based Python application. The application supports three roles: admin, professor, and student. Here, Admin manages records and user accounts whereas professors can see assigned-course information, and students can view their own grades and marks.

2. Lab 1 requirement mapping

Lab 1 requirement	How it is implemented	Status
Student records using linked list or array	Student records are loaded into LinkedList nodes through the Student class.	Met
Add, delete, modify student records	Handled by add_new_student(), delete_new_student(), and update_student_record().	Met
Search and sorted display	search_student() and sort_records() support search and sorting; tests print search and sort timing.	Met
Average/median or course statistics	course_stats() produces summary statistics for a course.	Met
Reports by course, professor, or student	ReportGenerator.display_grade_report() supports all three report types.	Met
CSV persistence for students, courses, professors, login	read_csv() and write_csv() maintain the four CSV files.	Met
Add, delete, modify courses	Course class provides add_new_course(), delete_new_course(), modify_course().	Met
Add, delete, modify professors	Professor class provides add_new_professor(), delete_professore(), modify_professor_details().	Met
Encrypted password storage	Passwords are saved in login.csv using a Caesar cipher with shift 13 (ROT13).	Met

3. Class design summary

Class / function group	Main responsibility	Key members
TextSecurity	Encrypts and decrypts stored passwords.	encrypt(), decrypt()
Session	Tracks the current user and role-based access.	login(), logout(), require()
Node / LinkedList	Stores student enrollment rows in memory.	append(), find(), find_all(), delete()
Grades	Maps marks to letter grades and supports grade reporting helpers.	marks_to_grade(), display_grade_report()
Student	Loads, updates, searches, sorts, and reports student enrollment records.	add_new_student(), delete_new_student(), update_student_record(), search_student(), sort_records(), course_stats()
Course	Maintains courses.csv.	add_new_course(), delete_new_course(), modify_course()
Professor	Maintains professors.csv and assigned-course details.	add_new_professor(), delete_professore(), modify_professor_details(), show_course_details_by_professor()
LoginUser / Admin	Handles login, registration, password change, and admin user management.	Login(), Change_password(), register(), add_user(), remove_user(), reset_password()
ReportGenerator / menus	Generates reports and connects all features through the console workflow.	display_grade_report(), startup_sync(), main_menu()

4. OOPS design

IS-A relationship: Admin inherits from LoginUser.

HAS-A relationships:

- Here Student has a LinkedList.
- LinkedList has Node objects
- ReportGenerator uses a Student object to build reports.

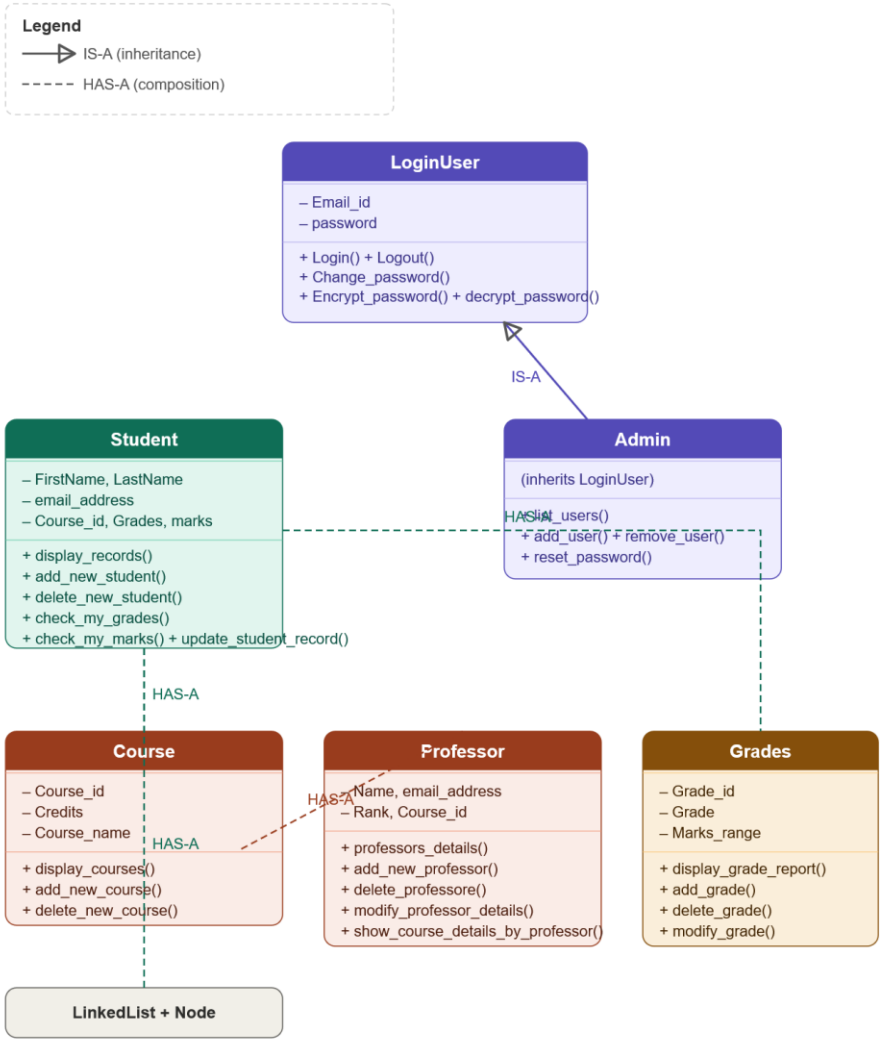

Figure 1. OOD relationship diagram included in the original report.

5. CSV files and actual records

The report below uses the current CSV contents from the project files, so the student, course, professor, and login records match the code base and dataset exactly.

5.1 students.csv

Email_address	First_name	Last_name	Course_id	Grade	Marks
sam@mycsu.edu	Sam	Carpenter	DATA200	A	96
sam@mycsu.edu	Sam	Carpenter	DATA220	B	85
sam@mycsu.edu	Sam	Carpenter	CS101	A	91
jane@mycsu.edu	Jane	Doe	DATA200	B	83
jane@mycsu.edu	Jane	Doe	DATA220	C	74

leo@mycsu.edu	Leo	Nguyen	CS101	C	74
priya@mycsu.edu	Priya	Sharma	DATA220	A	92
priya@mycsu.edu	Priya	Sharma	DATA200	A	95
raj@mycsu.edu	Raj	Patel	DATA200	C	78
raj@mycsu.edu	Raj	Patel	DATA220	B	88

5.2 courses.csv

Course_id	Course_name	Credits	Description
DATA200	Data Science Fundamentals	3	Provides insight about DS and Python
DATA220	Statistics	3	Applied statistics for data science
CS101	Intro to Computer Science	3	Foundations of computer science

5.3 professors.csv

Professor_id	Professor_name	Rank	Course_id
saini@sjsu.edu	Paramdeep Saini	Professor	DATA200
masum@sjsu.edu	Muhammad Masum	Professor	DATA220
john@sjsu.edu	John Smith	Professor	CS101

5.4 login.csv

We are keeping passwords in encrypted form. In this process:

- Admin123! will become Nqzva123!
- Prof123! will become Cebs123!
- Student123! becomes Fghqrag123! using ROT13

User_id	Password (encrypted)	Role
admin@mycsu.edu	Nqzva123!	admin
saini@sjsu.edu	Cebs123!	professor
masum@sjsu.edu	Cebs123!	professor
sam@mycsu.edu	Fghqrag123!	student
jane@mycsu.edu	Fghqrag123!	student
leo@mycsu.edu	Fghqrag123!	student

priya@mycsu.edu	Fghqrag123!	student
raj@mycsu.edu	Fghqrag123!	student
john@sjsu.edu	Cebs123!	professor

6. Code summary

- **CSV helpers:** read_csv() loads rows into dictionaries and write_csv() rewrites updated CSV files.
- **Security:** TextSecurity implements a Caesar cipher with shift 13 for password storage.
- **Access control:** Session keeps the logged-in role and protects admin/professor/student actions.
- **Student workflow:** Student handles loading, saving, CRUD operations, search, sort, grade lookup, and course statistics.
- **Flow of Course and professor:** Here the course as well as professor have their own CSV files and support add, delete, modify, and display operations.
- **Authentication and admin tools:** Here LoginUser manages login, logout, registration, and password changes; Admin extends it with list, add, remove, and reset functions.
- **Reporting and startup:** ReportGenerator builds reports by course, professor, and student, while startup_sync() aligns login/course data with student records when the application starts.

7. Unit testing summary

The current unit-test file contains 107 tests, and all tests pass with unittest. The suite checks the core Lab 1 workflows rather than claiming 100% line coverage.

Metric	**Result**
Total tests run	107
Tests passed	107
Tests failed	0
Search timing on 1000 records	0.5055 ms total | 0.0505 ms avg
Sort timing - email ascending	0.0102 ms
Sort timing - marks ascending	0.0114 ms
Sort timing - marks descending	0.0169 ms
Sort timing - name ascending	0.0063 ms
Test class	**Number of tests**
TestSession	6
TestStudent	14
TestCourse	5

TestProfessor	5
TestLoginUser	5
TestAdmin	5
TestStudentExtra	18
TestCourseExtra	5
TestProfessorExtra	5
TestAdminExtra	4

The tests verify student CRUD, multi-course enrollment, duplicate rejection, search and sorting, course and professor management, login and password behavior, admin operations, and several negative cases. They are strong evidence for the required Lab 1 features, although some interactive menu branches are still better demonstrated by the screenshots than by automated tests.

8. How to run

1. Keep checkmygrade.py, test_checkmygrade.py, and the four CSV files in the same folder.
2. Run the application with: python checkmygrade.py
3. Run the tests with: python -m unittest test_checkmygrade.py
4. Use the login.csv accounts to sign in as admin, professor, or student.

9. Program output screenshots

All screenshots from the original report are retained below as evidence of execution output. They are kept in the report and were not removed.

Logging in as Admin

```
PS C:\Users\58in\Downloads\PythonProfParamdeepSaini\LAB1> & C:/Users/58in/AppData/Local/Programs/Python/Python313/python.exe c:/Users/58in/D
ownloads/PythonProfParamdeepSaini/LAB1/checkmygrade.py
xe c:/Users/58in/Downloads/PythonProfParamdeepSaini/LAB1/checkmygrade.py

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   Not logged in
==============================================
  1. Login
  2. Register as student
  0. Exit

  Select: 1
  User ID: admin@mycsu.edu
  Password: Admin123!
  Login successful. Welcome admin@mycsu.edu! Role: admin

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   admin@mycsu.edu  [admin]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  5. User Management  [admin]
  6. Change Password
  7. Logout
  0. Exit

  Select: 1
```

Displaying all records of students by Admin:

Screenshot 1

```
-- Student Records --
1. Display  2. Add  3. Delete  4. Update
5. Search  6. Sort  8. Course Stats
0. Back
Select: 1
Email                        Name                 Course     Grade  Marks
----------------------------------------------------------------------
sam@mycsu.edu                Sam Carpenter        DATA200    A      96
sam@mycsu.edu                Sam Carpenter        DATA220    B      85
sam@mycsu.edu                Sam Carpenter        CS101      A      91
jane@mycsu.edu               Jane Doe             DATA200    B      83
jane@mycsu.edu               Jane Doe             DATA220    C      74
leo@mycsu.edu                Leo Nguyen           CS101      C      74
priya@mycsu.edu              Priya Sharma         DATA220    A      92
priya@mycsu.edu              Priya Sharma         DATA200    A      95
raj@mycsu.edu                Raj Patel            DATA200    C      78
raj@mycsu.edu                Raj Patel            DATA220    B      88

-- Student Records --
1. Display  2. Add  3. Delete  4. Update
5. Search  6. Sort  8. Course Stats
0. Back
Select: 2
Email: maria@mycsu.edu
First name: Maria
Last name: Lopez
Course ID: DATA220
Marks (0-100): 91
Student maria@mycsu.edu enrolled in DATA220.
Login created for maria@mycsu.edu (password: Student123!)

-- Student Records --
1. Display  2. Add  3. Delete  4. Update
```

Screenshot 2

Finding students by their email id

```
  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 1
  Email                        Name                 Course     Grade  Marks
  ---------------------------------------------------------------------------
  sam@mycsu.edu                Sam Carpenter        DATA200    A      96
  sam@mycsu.edu                Sam Carpenter        DATA220    B      85
  sam@mycsu.edu                Sam Carpenter        CS101      A      91
  jane@mycsu.edu               Jane Doe             DATA200    B      83
  jane@mycsu.edu               Jane Doe             DATA220    C      74
  leo@mycsu.edu                Leo Nguyen           CS101      C      74
  priya@mycsu.edu              Priya Sharma         DATA220    A      92
  priya@mycsu.edu              Priya Sharma         DATA200    A      95
  raj@mycsu.edu                Raj Patel            DATA200    C      78
  raj@mycsu.edu                Raj Patel            DATA220    B      88
  maria@mycsu.edu              Maria Lopez          DATA220    A      91

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 5
  Email: maria@mycsu.edu
  Search time: 0.0211 ms
  Found: {'Email_address': 'maria@mycsu.edu', 'First_name': 'Maria', 'Last_name': 'Lopez', 'Course_id': 'DATA220', 'Grade': 'A', 'Marks': '9
1'}

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 6
```

Screenshot 3

Sorting student records by Admin

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python

  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 6
  Sort by: 1.Marks  2.Email  3.Name
  Select: 1
  Descending? (y/n): y
  Sort time: 0.0818 ms
  sam@mycsu.edu                DATA200    Marks: 96
  priya@mycsu.edu              DATA200    Marks: 95
  priya@mycsu.edu              DATA220    Marks: 92
  sam@mycsu.edu                CS101      Marks: 91
  maria@mycsu.edu              DATA220    Marks: 91
  raj@mycsu.edu                DATA220    Marks: 88
  sam@mycsu.edu                DATA220    Marks: 85
  jane@mycsu.edu               DATA200    Marks: 83
  raj@mycsu.edu                DATA200    Marks: 78
  jane@mycsu.edu               DATA220    Marks: 74
  leo@mycsu.edu                CS101      Marks: 74

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   admin@mycsu.edu  [admin]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
```

Screenshot 4

Getting overall reports, their median and average marks by COURSE ID

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   Python

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   admin@mycsu.edu  [admin]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  5. User Management  [admin]
  6. Change Password
  7. Logout
  0. Exit

  Select: 4

  -- Reports --
  1. By Course  2. By Professor  3. By Student
  0. Back
  Select: 1
  Course ID: DATA200

  === Grade Report: Course DATA200 ===
  sam@mycsu.edu               Sam Carpenter        DATA200    Grade: A  Marks: 96
  jane@mycsu.edu              Jane Doe             DATA200    Grade: B  Marks: 83
  priya@mycsu.edu             Priya Sharma         DATA200    Grade: A  Marks: 95
  raj@mycsu.edu               Raj Patel            DATA200    Grade: C  Marks: 78
  DATA200 | Students: 4 | Avg: 88.00 | Median: 89.00
```

Screenshot 5

Logging out and then logging in as a professor

```
  -- Reports --
  1. By Course  2. By Professor  3. By Student
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   admin@mycsu.edu  [admin]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  5. User Management  [admin]
  6. Change Password
  7. Logout
  0. Exit

  Select: 7
  admin@mycsu.edu logged out.

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   Not logged in
==============================================
  1. Login
  2. Register as student
  0. Exit

  Select: 1
  User ID: saini@sjsu.edu
  Password: Prof123!
  Login successful. Welcome saini@sjsu.edu! Role: professor
```

Screenshot 6

Displaying students under professor

```
  Login successful. Welcome saini@sjsu.edu! Role: professor

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   saini@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 1

  -- Student Records --
  1. Display
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 1
  Email                        Name                 Course     Grade  Marks
  ---------------------------------------------------------------------
  sam@mycsu.edu                Sam Carpenter        DATA200    A      96
  jane@mycsu.edu               Jane Doe             DATA200    B      83
  priya@mycsu.edu              Priya Sharma         DATA200    A      95
  raj@mycsu.edu                Raj Patel            DATA200    C      78

  -- Student Records --
  1. Display
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 0
```

Screenshot 7

Checking subject details under that professor

```
==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   saini@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 2

  -- Courses --
  1. Display
  0. Back
  Select: 1
  ID          Name                     Credits  Description
  ---------------------------------------------------------------
  DATA200     Data Science Fundamentals 3       Provides insight about DS and Python

  -- Courses --
  1. Display
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   saini@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
```

Screenshot 8

Displaying all the students with their marks and grades under that professor

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 3

  -- Professors --
  1. Display
  0. Back
  Select: 1
  saini@sjsu.edu              Paramdeep Saini        Professor          DATA200

  -- Professors --
  1. Display
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   saini@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit
```

Screenshot 9

```
  Select: 4

  -- Reports --
  1. By Course  2. By Professor  3. By Student
  0. Back
  Select: 1
  Course ID: DATA200

  === Grade Report: Course DATA200 ===
  sam@mycsu.edu               Sam Carpenter        DATA200    Grade: A  Marks: 96
  jane@mycsu.edu              Jane Doe             DATA200    Grade: B  Marks: 83
  priya@mycsu.edu             Priya Sharma         DATA200    Grade: A  Marks: 95
  raj@mycsu.edu               Raj Patel            DATA200    Grade: C  Marks: 78
  DATA200 | Students: 4 | Avg: 88.00 | Median: 89.00

  -- Reports --
  1. By Course  2. By Professor  3. By Student
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   saini@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 7
```

Screenshot 10

Logging in as another professor and displaying the students under him:

```
  Select: 7
  saini@sjsu.edu logged out.

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   Not logged in
==============================================
  1. Login
  2. Register as student
  0. Exit

  Select: 1
  User ID: masum@sjsu.edu
  Password: Prof123!
  Login successful. Welcome masum@sjsu.edu! Role: professor

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   masum@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 1

  -- Student Records --
  1. Display
  5. Search  6. Sort  8. Course Stats
```

Screenshot 11

```
   Select: 1

   -- Student Records --
   1. Display
   5. Search  6. Sort  8. Course Stats
   0. Back
   Select: 1
   Email                       Name                Course     Grade  Marks
   ---------------------------------------------------------------------
   sam@mycsu.edu               Sam Carpenter       DATA220    B      85
   jane@mycsu.edu              Jane Doe            DATA220    C      74
   priya@mycsu.edu             Priya Sharma        DATA220    A      92
   raj@mycsu.edu               Raj Patel           DATA220    B      88
   maria@mycsu.edu             Maria Lopez         DATA220    A      91

   -- Student Records --
   1. Display
   5. Search  6. Sort  8. Course Stats
   0. Back
   Select: 0

  ==============================================
    CheckMyGrade  |  DATA 200  |  SJSU
    masum@sjsu.edu  [professor]
  ==============================================
   1. Student Records
   2. Course Management
   3. Professor Management
   4. Reports
   6. Change Password
   7. Logout
   0. Exit

   Select: 2
```

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Screenshot 12

Checking other Professor details

```
  Select: 2

  -- Courses --
  1. Display
  0. Back
  Select: 1
  ID          Name                    Credits  Description
  ----------------------------------------------------------------
  DATA220     Statistics              3        Applied statistics for data science

  -- Courses --
  1. Display
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   masum@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 3

  -- Professors --
  1. Display
  0. Back
  Select: 1
  masum@sjsu.edu              Muhammad Masum         Professor          DATA220
```

Screenshot 13

Logging in as a student:

```
  masum@sjsu.edu              Muhammad Masum         Professor          DATA220

  -- Professors --
  1. Display
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   masum@sjsu.edu  [professor]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 0
  Goodbye!
PS C:\Users\58in\Downloads\PythonProfParamdeepSaini\LAB1> & C:/Users/58in/AppData/Local/Programs/Python/Python313/python.exe c:/Users/58in/D
ownloads/PythonProfParamdeepSaini/LAB1/checkmygrade.py

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   Not logged in
==============================================
  1. Login
  2. Register as student
  0. Exit

  Select: 1
  User ID: maria@mycsu.edu
```

Screenshot 14

```
  Password: Student123!
  Login successful. Welcome maria@mycsu.edu! Role: student

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   maria@mycsu.edu  [student]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 1

  -- Student Records --
  9. My Grades & Marks
  0. Back
  Select: 9
  Grades for maria@mycsu.edu:
    DATA220      Grade: A  Marks: 91
  Marks for maria@mycsu.edu:
    DATA220      Marks: 91

  -- Student Records --
  9. My Grades & Marks
  0. Back
  Select: 0

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   maria@mycsu.edu  [student]
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF { } Python 3.13.1 Python 3.13 (64-

Screenshot 15

Checking grades of the student

```
==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   maria@mycsu.edu  [student]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  6. Change Password
  7. Logout
  0. Exit

  Select: 7
  maria@mycsu.edu logged out.

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   Not logged in
==============================================
  1. Login
  2. Register as student
  0. Exit

  Select:
```

Screenshot 16

Logging in as an admin-

```
  Password: Admin123!
  Login successful. Welcome admin@mycsu.edu! Role: admin

==============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   admin@mycsu.edu  [admin]
==============================================
  1. Student Records
  2. Course Management
  3. Professor Management
  4. Reports
  5. User Management  [admin]
  6. Change Password
  7. Logout
  0. Exit

  Select: 1

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 1
  Email                        Name                 Course     Grade  Marks
  ---------------------------------------------------------------------------
  sam@mycsu.edu                Sam Carpenter        DATA200    A      96
  sam@mycsu.edu                Sam Carpenter        DATA220    B      85
  sam@mycsu.edu                Sam Carpenter        CS101      A      91
  jane@mycsu.edu               Jane Doe             DATA200    B      83
  jane@mycsu.edu               Jane Doe             DATA220    C      74
  leo@mycsu.edu                Leo Nguyen           CS101      C      74
  priya@mycsu.edu              Priya Sharma         DATA220    A      92
  priya@mycsu.edu              Priya Sharma         DATA200    A      95
  raj@mycsu.edu                Raj Patel            DATA200    C      78
```

Screenshot 17

Deleting a student by admin:

```
  raj@mycsu.edu                Raj Patel            DATA200    C      78
  raj@mycsu.edu                Raj Patel            DATA220    B      88
  maria@mycsu.edu              Maria Lopez          DATA220    A      91

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 3
  Email: maria@mycsu.edu
  Course ID (blank = delete all): DATA220
  Removed maria@mycsu.edu from DATA220.
  Login removed for maria@mycsu.edu.

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
  5. Search  6. Sort  8. Course Stats
  0. Back
  Select: 1
  Email                        Name                 Course     Grade  Marks
  ---------------------------------------------------------------------------
  sam@mycsu.edu                Sam Carpenter        DATA200    A      96
  sam@mycsu.edu                Sam Carpenter        DATA220    B      85
  sam@mycsu.edu                Sam Carpenter        CS101      A      91
  jane@mycsu.edu               Jane Doe             DATA200    B      83
  jane@mycsu.edu               Jane Doe             DATA220    C      74
  leo@mycsu.edu                Leo Nguyen           CS101      C      74
  priya@mycsu.edu              Priya Sharma         DATA220    A      92
  priya@mycsu.edu              Priya Sharma         DATA200    A      95
  raj@mycsu.edu                Raj Patel            DATA200    C      78
  raj@mycsu.edu                Raj Patel            DATA220    B      88

  -- Student Records --
  1. Display  2. Add  3. Delete  4. Update
```

Screenshot 18

Register as a student

```
PS C:\Users\58in\Downloads\PythonProfParamdeepSaini\LAB1> & C:/Users/58in/AppData/Local/Programs/Python/Python313/python.exe c:
ers/58in/Downloads/PythonProfParamdeepSaini/LAB1/checkmygrade.py

=============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   Not logged in
=============================================
  1. Login
  2. Register as student
  0. Exit

  Select: 2
  Email: prash@csu.com
  Password: Student123!
  User prash@csu.com registered as student.

=============================================
   CheckMyGrade  |  DATA 200  |  SJSU
   Not logged in
=============================================
  1. Login
  2. Register as student
  0. Exit
```

Screenshot 19

Update the existing record

```
sam@mycsu.edu              Sam Carpenter        DATA200    A      96
sam@mycsu.edu              Sam Carpenter        DATA220    B      85
sam@mycsu.edu              Sam Carpenter        CS101      A      91
jane@mycsu.edu             Jane Doe             DATA200    B      83
jane@mycsu.edu             Jane Doe             DATA220    C      74
leo@mycsu.edu              Leo Nguyen           CS101      C      74
priya@mycsu.edu            Priya Sharma         DATA220    A      92
priya@mycsu.edu            Priya Sharma         DATA200    A      95
raj@mycsu.edu              Raj Patel            DATA200    C      78
raj@mycsu.edu              Raj Patel            DATA220    B      88

-- Student Records --
1. Display  2. Add  3. Delete  4. Update
5. Search  6. Sort  8. Course Stats
0. Back
Select: 4
Email: raj@mycsu.edu
Field (First_name/Last_name/Course_id/Marks): Marks
New value: 99
Updated Marks for raj@mycsu.edu.

-- Student Records --
1. Display  2. Add  3. Delete  4. Update
5. Search  6. Sort  8. Course Stats
0. Back
Select:
```

Screenshot 12

Output of Unit Testing

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
  [sort] email ascending: 0.0366 ms
ok
test_sort_marks_ascending_timed (__main__.TestStudent.test_sort_marks_ascending_timed) ...   Student vmutm@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for vmutm@test.edu (password: Student123!)
  Student qyuqa@test.edu enrolled in DATA200.
  Login created for qyuqa@test.edu (password: Student123!)
  Student gzzpy@test.edu enrolled in DATA200.
  Login created for gzzpy@test.edu (password: Student123!)
  Student wqlgs@test.edu enrolled in DATA200.
  Login created for wqlgs@test.edu (password: Student123!)

  [sort] marks ascending: 0.0282 ms
ok
test_sort_marks_descending_timed (__main__.TestStudent.test_sort_marks_descending_timed) ...   Student wfqfg@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for wfqfg@test.edu (password: Student123!)
  Student yqqpd@test.edu enrolled in DATA200.
  Login created for yqqpd@test.edu (password: Student123!)
  Student faspp@test.edu enrolled in DATA200.
  Login created for faspp@test.edu (password: Student123!)
  Student qwqbx@test.edu enrolled in DATA200.
  Login created for qwqbx@test.edu (password: Student123!)

  [sort] marks descending: 0.0220 ms
ok
test_sort_name_timed (__main__.TestStudent.test_sort_name_timed) ...   Student zara@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for zara@test.edu (password: Student123!)
  Student alice@test.edu enrolled in DATA200.
  Login created for alice@test.edu (password: Student123!)
  Student mike@test.edu enrolled in DATA200.
  Login created for mike@test.edu (password: Student123!)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
  Updated Marks for e@test.edu.
ok
test_auto_sync_course_created_on_student_add (__main__.TestStudentExtra.test_auto_sync_course_created_on_student_add) ...   Student nc@test.
edu enrolled in NEWCOURSE.
  Course NEWCOURSE auto-created in courses.csv.
  Login created for nc@test.edu (password: Student123!)
ok
test_auto_sync_login_created_on_student_add (__main__.TestStudentExtra.test_auto_sync_login_created_on_student_add) ...   Student nl@test.ed
u enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for nl@test.edu (password: Student123!)
ok
test_auto_sync_login_kept_if_other_enrollment (__main__.TestStudentExtra.test_auto_sync_login_kept_if_other_enrollment) ...   Student ke@tes
t.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for ke@test.edu (password: Student123!)
  Student ke@test.edu enrolled in DATA220.
  Course DATA220 auto-created in courses.csv.
  Removed ke@test.edu from DATA200.
ok
test_auto_sync_login_removed_on_student_delete (__main__.TestStudentExtra.test_auto_sync_login_removed_on_student_delete) ...   Student rd@t
est.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for rd@test.edu (password: Student123!)
  All enrollments for rd@test.edu deleted.
  Login removed for rd@test.edu.
ok
test_check_my_marks (__main__.TestStudentExtra.test_check_my_marks) ...   Student m@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for m@test.edu (password: Student123!)
  Marks for m@test.edu:
    DATA200      Marks: 92
ok
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

  Login created for tulht@test.edu (password: Student123!)
  Student qjnqj@test.edu enrolled in DATA200.
  Login created for qjnqj@test.edu (password: Student123!)
ok
test_course_stats_no_records (__main__.TestStudentExtra.test_course_stats_no_records) ... ok
test_display_records_professor_scoped (__main__.TestStudentExtra.test_display_records_professor_scoped) ...   Student sc@test.edu enrolled i
n DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for sc@test.edu (password: Student123!)
  Student sc2@test.edu enrolled in DATA220.
  Course DATA220 auto-created in courses.csv.
  Login created for sc2@test.edu (password: Student123!)
ok
test_get_all_returns_list (__main__.TestStudentExtra.test_get_all_returns_list) ...   Student g@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for g@test.edu (password: Student123!)
ok
test_grades_invalid_marks (__main__.TestStudentExtra.test_grades_invalid_marks) ...   Invalid marks 'abc'.
ok
test_grades_marks_to_grade_all_ranges (__main__.TestStudentExtra.test_grades_marks_to_grade_all_ranges) ... ok
test_linked_list_delete_specific (__main__.TestStudentExtra.test_linked_list_delete_specific) ...   Student dl@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for dl@test.edu (password: Student123!)
  Student dl@test.edu enrolled in DATA220.
  Course DATA220 auto-created in courses.csv.
ok
test_linked_list_find_all (__main__.TestStudentExtra.test_linked_list_find_all) ...   Student ll@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for ll@test.edu (password: Student123!)
  Student ll@test.edu enrolled in DATA220.
  Course DATA220 auto-created in courses.csv.
ok
test_search_not_found (__main__.TestStudentExtra.test_search_not_found) ...   Search time: 0.0045 ms
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

  Search time: 0.0112 ms
  Found: {'Email_address': 's@test.edu', 'First_name': 'S', 'Last_name': 'T', 'Course_id': 'DATA200', 'Grade': 'A', 'Marks': '90'}
ok
test_student_cannot_see_others_marks (__main__.TestStudentExtra.test_student_cannot_see_others_marks) ...   Student o@test.edu enrolled in D
ATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for o@test.edu (password: Student123!)
ok
test_update_first_name (__main__.TestStudentExtra.test_update_first_name) ...   Student upd@test.edu enrolled in DATA200.
  Course DATA200 auto-created in courses.csv.
  Login created for upd@test.edu (password: Student123!)
  Updated First_name for upd@test.edu.
ok
test_update_nonexistent_student (__main__.TestStudentExtra.test_update_nonexistent_student) ...   Student ghost@test.edu not found.
ok
test_add_new_student_and_validations (__main__.TestStudentFull.test_add_new_student_and_validations) ...   Student new@mycsu.edu enrolled in
 DATA300.
  Course DATA300 auto-created in courses.csv.
  Login created for new@mycsu.edu (password: Student123!)
  new@mycsu.edu already enrolled in DATA300.
  Error: All fields required.
  Error: Marks must be numeric.
  Access denied. Required: admin. Your role: student.
ok
test_check_my_grades_and_marks_access (__main__.TestStudentFull.test_check_my_grades_and_marks_access) ... ok
test_delete_new_student_specific_and_all_and_not_found (__main__.TestStudentFull.test_delete_new_student_specific_and_all_and_not_found) ..
   Removed sam@mycsu.edu from CS101.
  All enrollments for alex@mycsu.edu deleted.
  Student ghost@mycsu.edu not found.
  Error: Email required.
  Access denied. Required: admin. Your role: student.
ok
test_display_records_admin_and_professor_filter (__main__.TestStudentFull.test_display_records_admin_and_professor_filter) ... ok
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

test_display_records_denied_or_empty (__main__.TestStudentFull.test_display_records_denied_or_empty) ... ok
test_load_and_save (__main__.TestStudentFull.test_load_and_save) ... ok
test_search_sort_stats_and_get_all (__main__.TestStudentFull.test_search_sort_stats_and_get_all) ...   Search time: 0.0097 ms
  Found: {'Email_address': 'alex@mycsu.edu', 'First_name': 'Alex', 'Last_name': 'Stone', 'Course_id': 'DATA200', 'Grade': 'C', 'Marks': '74'
}
  Search time: 0.0166 ms
  Not found.
  Access denied. Required: admin, professor. Your role: student.
ok
test_sync_course_add_and_no_duplicate (__main__.TestStudentFull.test_sync_course_add_and_no_duplicate) ...   Course NEW123 auto-created in c
ourses.csv.
ok
test_sync_login_add_and_no_duplicate (__main__.TestStudentFull.test_sync_login_add_and_no_duplicate) ...   Login created for alex2@mycsu.edu
 (password: Student123!)
ok
test_sync_login_remove_only_when_no_enrollments_left (__main__.TestStudentFull.test_sync_login_remove_only_when_no_enrollments_left) ...   A
ll enrollments for sam@mycsu.edu deleted.
  Login removed for sam@mycsu.edu.
ok
test_update_student_record_and_marks_regrade (__main__.TestStudentFull.test_update_student_record_and_marks_regrade) ...   Updated Marks for
 alex@mycsu.edu.
  Error: Marks must be numeric.
  Student ghost@mycsu.edu not found.
  Access denied. Required: admin. Your role: student.
ok

----------------------------------------------------------------------
Ran 107 tests in 1.596s

OK
PS C:\Users\58in\Downloads\PythonProfParamdeepSaini\LAB1>
```

10. Conclusion

The rewritten report is aligned to the actual code, current CSV records, and present unit-test file. It keeps the required screenshots, corrects the dataset descriptions, and stays focused on the features expected in Lab 1.